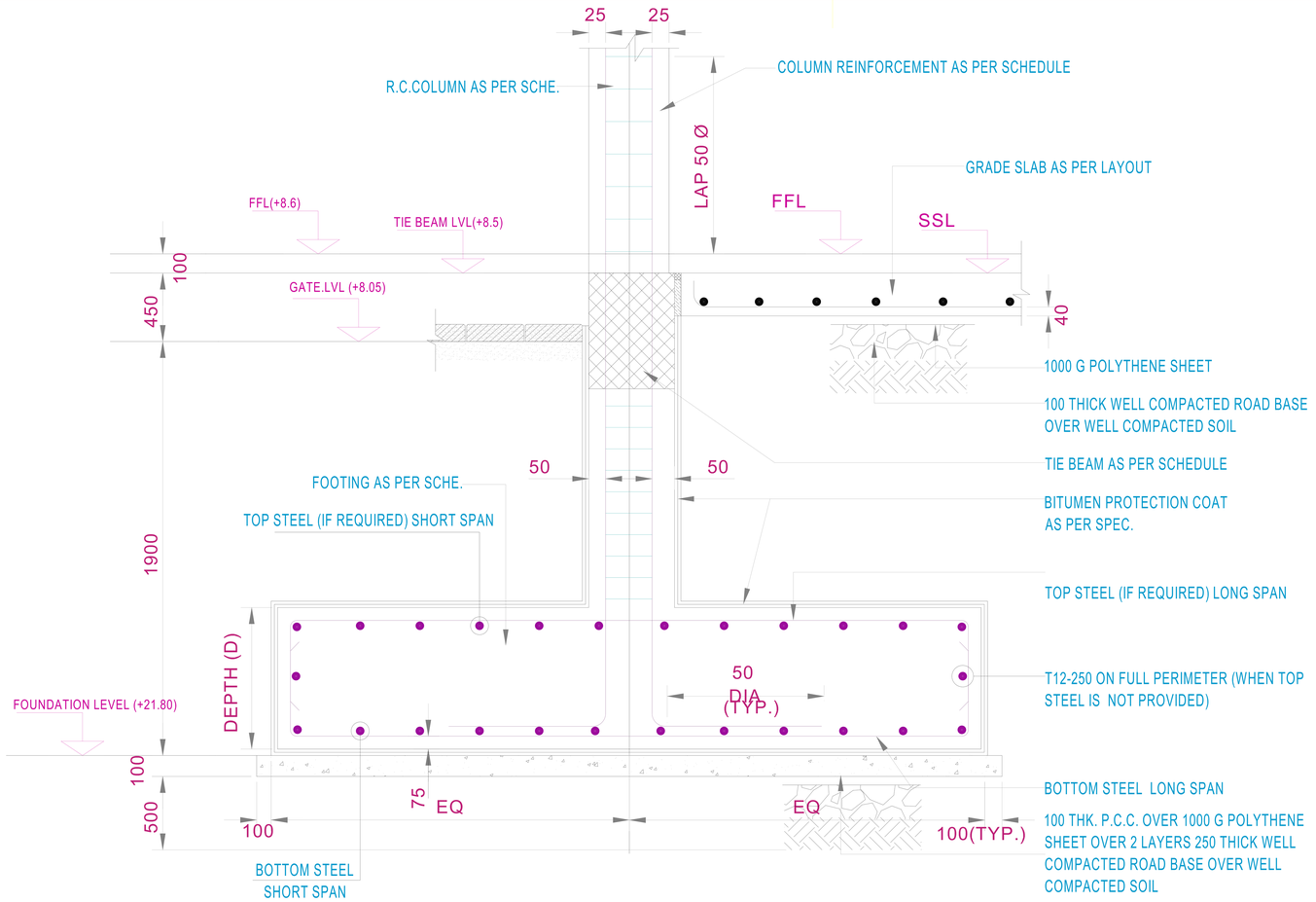


(TYPICAL SECTION FOR FOOTING)

ALLOWABLE BEARING CAPACITY FOR ISOLATED/ STRIP FOOTINGS

Width of Footing	Allowable Bearing Pressure	Modulus of sub-grade
Up to 2.00m	140 kN/m2	16,800 kN/m3
Up to 3.00m	130 kN/m2	15,600 kN/m3
Up to 4.00m	120 kN/m2	14,400 kN/m3
Up to 5.00m&wider	110 kN/m2	13,200 kN/m3

- Excavate and level the existing soil at the Level of -2.50m below the D.LVL level established on the the adjacent unpaved road .

-Compact using heavy rollers, the soil at the bottom of the excavation at a level of 2.50m from the adjacent unpaved road to at least 95% of its maximum dry density at its optimum moisture content obtained from modified proctor compaction test.

- Place tow (2) layer of selected granular backfill materials (Road Base) 25cm thick to reach the required foundation level. the layer shall be compacted by heavy vibratory roller to a degree of compaction not less than 98% of the maximum dry density as obtained by modified proctor compaction test (BS 1377:1990 PART 4) .
-
Plate load test must be carried out at the foundation level to verify the required allowable bearing
capacity and expected settlement criteria.

- A 100mm thick layer of blinding concrete (PCC) should be placed to protect the compacted surface .

- Shallow foundation can be placed on engineered fill. (i.e., Approximately -1.40m below the D.LVL level established on the the adjacent unpaved road) .

DETAILS FONDATION

Govemment Of Ras Al Khaimah
R.A.K. Municipality
Buildings Administration

حكومة رأس الخيمة
دائرة البلدية
إدارة المباني

Approved On — اعتمد في

28 January 2026

Building Permit #: PRMT-26-0000880

Appoved By:

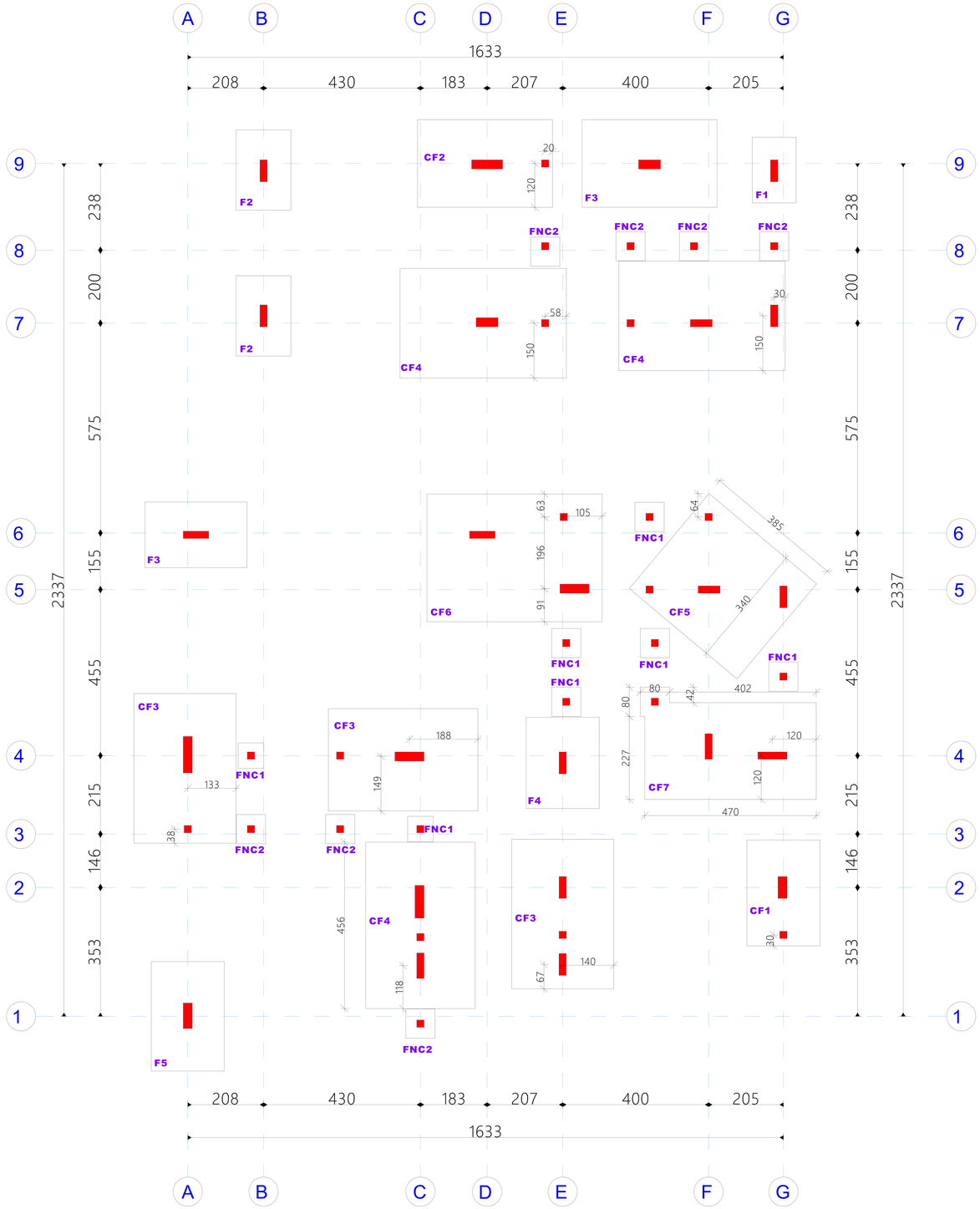
STRUCTURAL NOTES	CONCRETE COVERS :
ACI 318-11	Foundations cover = 75mm
F'C/FCU = 28/35 N/mm2	Columns cover = 40mm
FY = 460N/mm2	Walls cover = 30mm
FYt = 420N/mm2	Beams cover = 40mm
	Slabs & Stairs cover = 25mm

Rooms	Space	Number
Residential		
Commercial		
Industrial		
Goverment		
Resident.-Comm.		
Total		

Savana
Engineering Consultancy & Interior design

Client	MAYED MEERZA BASHEER MOHAMMED		
Project	DOUBLE STORY VILLA		
Drg.Title	DETAILS FOUNDATION		
Area	AWAFI	Floortype	
Case ID.	3136005	Draw. Type.	STRC
Sec No.	1	Com. No.	
Block No.		Plot No.	406131530
Bldg. No.	BLD-26-00570	Floor No.	
Date	1/2026	Sht. No.	S01
Scale	1/100	Job No.	

G.Manager	Architect	Designed By
ENG AYMEN	ENG IMEN	ENG IMEN



FONDATION LAYOUT

Govemment Of Ras Al Khaimah
R.A.K. Municipality
Buildings Administration

حكومة رأس الخيمة
دائرة البلدية
إدارة المباني

Approved On — اعتمد في

28 January 2026

Building Permit #: PRMT-26-0000880

Appoved By:

STRUCTURAL NOTES

ACI 318-11
F'C/FCU = 28/35 N/mm2
FY = 460N/mm2
FYt = 420N/mm2

CONCRETE COVERS :

Foundations cover = 75mm
Columns cover = 40mm
Walls cover = 30mm
Beams cover = 40mm
Slabs & Stairs cover = 25mm

Rooms	Space	Number
Residential		
Commercial		
Industrial		
Goverment		
Resident.-Comm.		
Total		

Savana
Engineering Consultancy & interior design

Client	MAYED MEERZA BASHEER MOHAMMED		
Project	DOUBLE STORY VILLA		
Drg.Title	FOUNDATION LAYOUT		
Area	AWAFI	Floortype	
Case ID.	3136005	Draw. Type.	STRC
Sec No.	1	Com. No.	
Block No.		Plot No.	406131530
Bldg. No.	BLD-26-00570	Floor No.	
Date	1/2026	Sht. No.	S02
Scale	1/100	Job No.	

G.Manager	Architect	Designed By
ENG AYMEN	ENG IMEN	ENG IMEN

TYPE	SIZE			REINFORCEMENT DETAILS				REMARKS.
	WIDTH (mm)	LENGTH (mm)	DEPTH (mm)	BOTTOM MESH		TOP MESH		
				SHORT BAR	LONG BAR	SHORT BAR	LONG BAR	
F1	1200	1800	400	T12-200	T12-200	2T12	2T12	
F2	1500	2200	400	T14-200	T14-200	3T12	3T12	
F3	1800	2800	400	T14-200	T14-200	3T12	3T12	
F4	2000	2500	400	T14-125	T14-125	3T12	3T12	
F5	2000	3000	400	T14-100	T14-100	3T12	3T12	
CF1	2000	2900	400	T14-125	T14-125	T12/250	T12/250	
CF2	2400	3700	400	T16-100	T16-100	T12/200	T12/200	
CF3	2800	4100	450	T16-150	T16-150	T12/250	T12/250	
CF4	3000	4560	550	T18-150	T18-150	T12/200	T12/200	
CF5	3400	3850	500	T16-150	T16-150	T12/200	T12/200	
CF6	3500	4800	450	T16-100	T16-100	T14/150	T14/150	
CF7	AS DRAWING		400	T16-150	T16-150	T12/250	T12/250	
FNC1	700	700	350	T12-250	T12-250	2T12	2T12	
FNC2	800	800	350	T12-250	T12-250	2T12	2T12	

Govemment Of Ras Al Khaimah
R.A.K. Municipality
Buildings Administration

حكومة رأس الخيمة
دائرة البلدية
إدارة المباني

Approved On — اعتمد في

28 January 2026

Building Permit #: PRMT-26-0000880

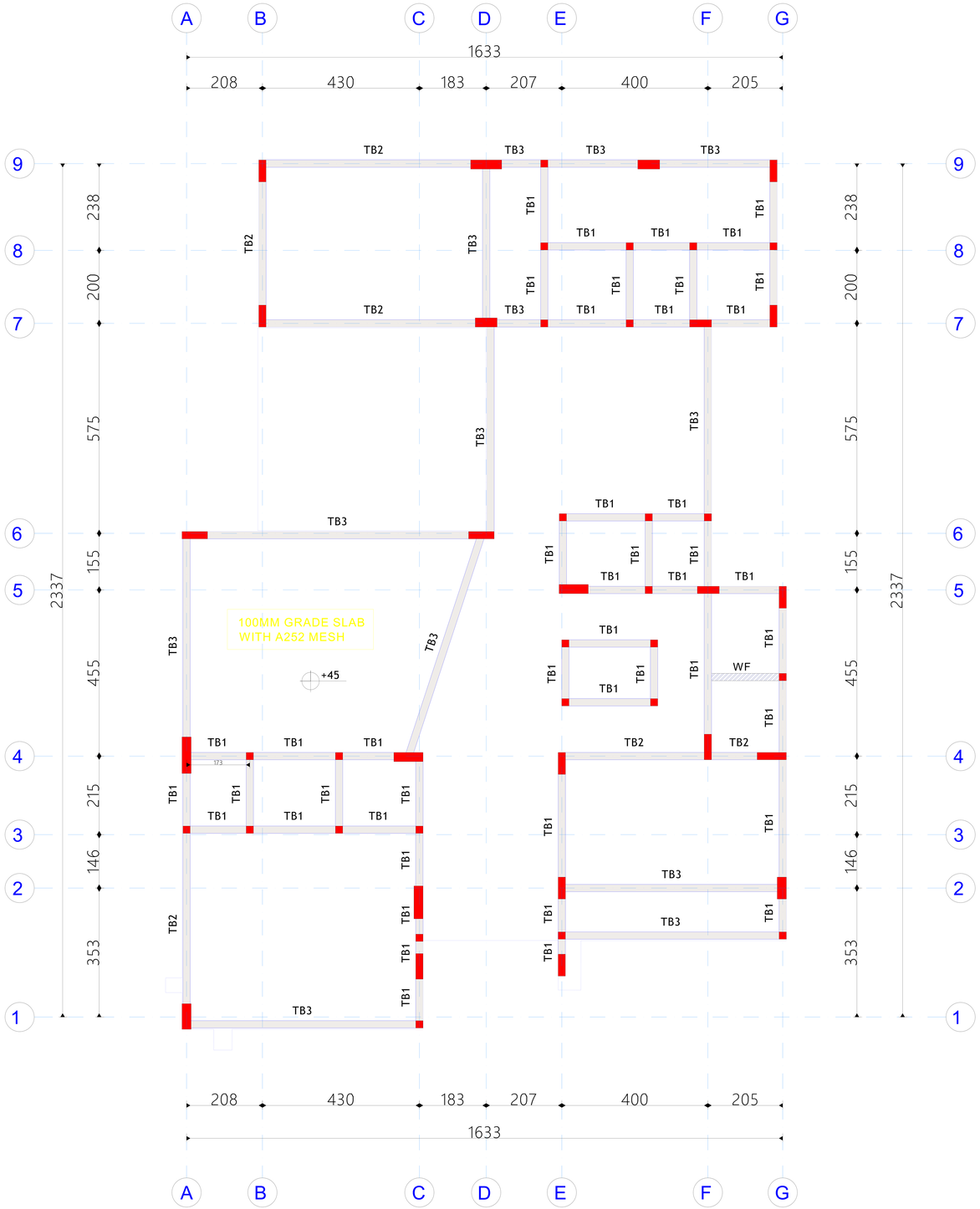
Appoved By:

Savana
Engineering Consultancy & interior design

Client	MAYED MEERZA BASHEER MOHAMMED		
Project	DOUBLE STORY VILLA		
Drg.Title	FOUNDATION SCHEDULE		
Area	AWAFI	Floortype	
Case ID.	3136005	Draw. Type.	STRC
Sec No.	1	Com. No.	
Block No.		Plot No.	406131530
Bldg. No.	BLD-26-00570	Floor No.	
Date	1/2026	Sht. No.	S03
Scale	1/100	Job No.	

G.Manager	Architect	Designed By
ENG AYMEN	ENG IMEN	ENG IMEN

SCHEDUEL FONDATION

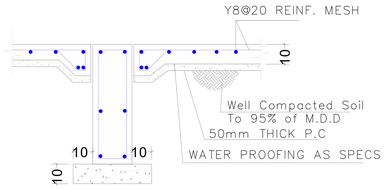


BEAM SIZE BOTTOM REINFORCEMENT				
TYPE	MM XMM	STRAIGHT (A)	EXTRA @ MID SPAN(B)	STRAIGHT (C)
TB1	200 X 500	3T14	----	3T14
TB2	200 X 600	4T14	----	4T14
TB3	200 X 700	4T14	2T14	4T14

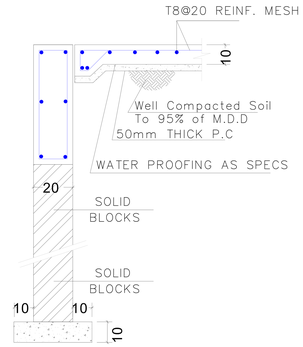
TOP REINFORCEMENT			
SIDE BARS O.E.F (F) EACH SIDE	EXTRA @ DISCONT. SUPPORT (E)	EXTRA @ CONT. SUPPORT (D)	NO OF LEGS
1T12	----	----	2
1T12	----	----	2
2T12	2T14	2T14	2

STIRRUPS	
AT SUPPORT	AT MIDSPAN
T8-150	T8-200
T8-150	T8-200
T8-150	T8-200

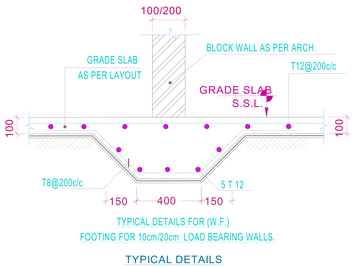
SCHEDULE OF GROUND BEAMS



TYPICAL INTERNAL GROUND BEAM
SC. (1/25)

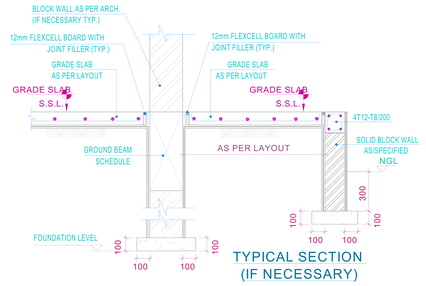


TYPICAL EXTERNAL GROUND BEAM
SC. (1/25)



TYPICAL DETAILS FOR (W.F.)
FOOTING FOR 10cm/20cm LOAD BEARING WALLS.

TYPICAL DETAILS



TYPICAL SECTION
(IF NECESSARY)

TIE BEAM LAYOUT

Govemment Of Ras Al Khaimah
R.A.K. Municipality
Buildings Administration

حكومة رأس الخيمة
دائرة البلدية
إدارة المباني

Approved On — اعتمد في

28 January 2026

Building Permit #: PRMT-26-0000880

Appoved By:

STRUCTURAL NOTES	CONCRETE COVERS :
ACI 318-11	Foundations cover = 75mm
F'C/FCU = 28/35 N/mm2	Columns cover = 40mm
FY = 460N/mm2	Walls cover = 30mm
FYt = 420N/mm2	Beams cover = 40mm
	Slabs & Stairs cover = 25mm

LEGEND:

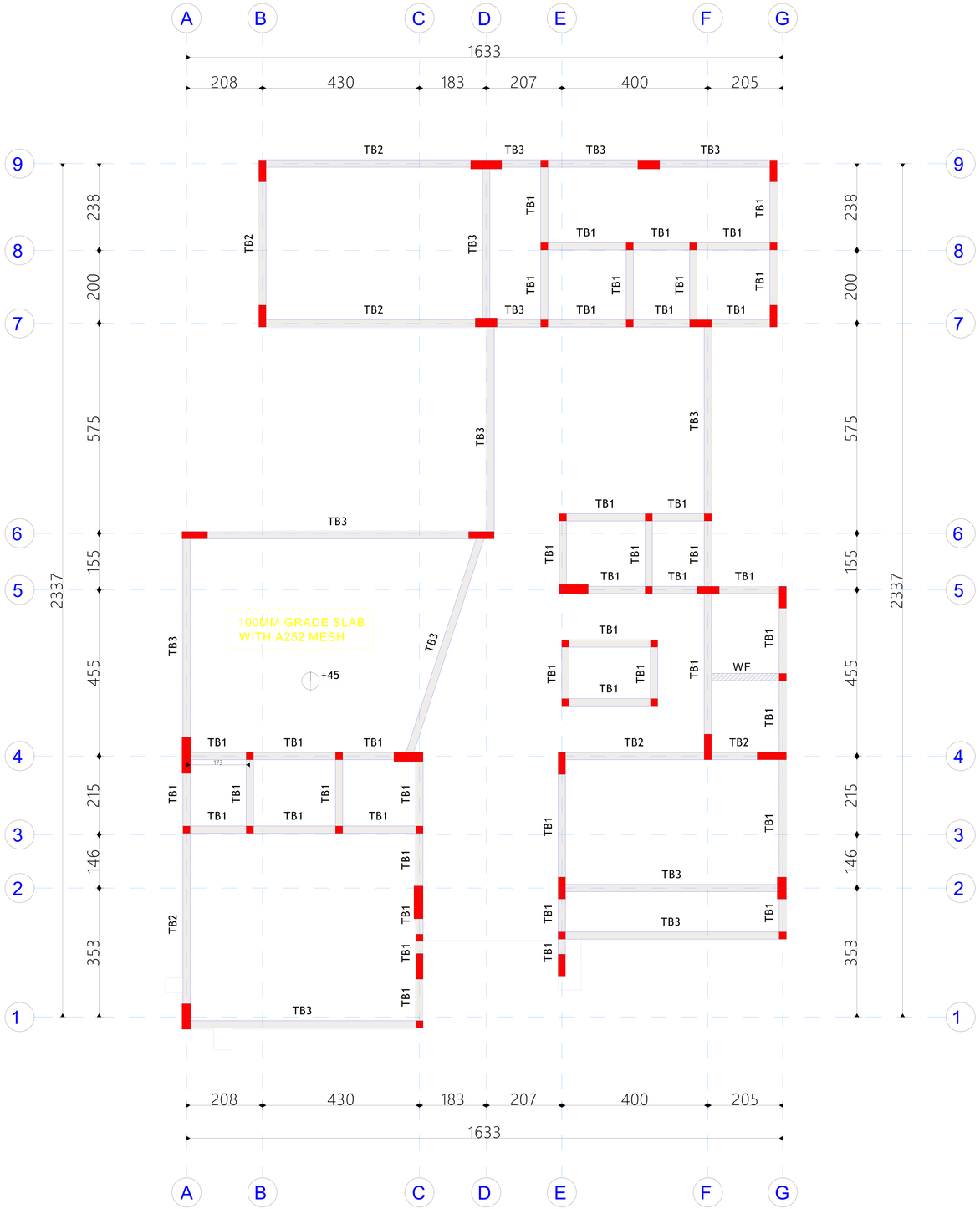
1) COLUMN / WALL TO BE CONTINUED

2) COLUMN / WALL TO BE TERMINATED

Savana
Engineering Consultancy & interior design

Client	MAYED MEERZA BASHEER MOHAMMED		
Project	DOUBLE STORY VILLA		
Drg.Title	TIE BEAM LAYOUT		
Area	AWAFI	Floortype	
Case ID.	3136005	Draw. Type.	STRC
Sec No.	1	Com. No.	
Block No.		Plot No.	406131530
Bldg. No.	BLD-26-00570	Floor No.	
Date	1/2026	Sht. No.	S04
Scale	1/100	Job No.	

G.Manager	Architect	Designed By
ENG AYMEN	ENG IMEN	ENG IMEN

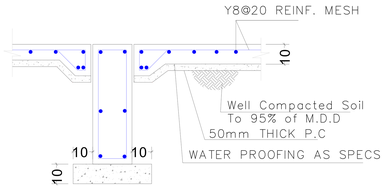


BEAM SIZE BOTTOM REINFORCEMENT				
TYPE	MM XMM	STRAIGHT (A)	EXTRA @ MID SPAN(B)	STRAIGHT (C)
TB1	200 X 500	3T14	----	3T14
TB2	200 X 600	4T14	----	4T14
TB3	200 X 700	4T14	2T14	4T14

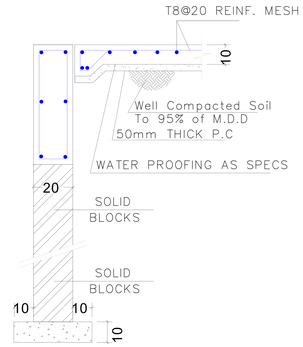
TOP REINFORCEMENT			
SIDE BARS O.E.F (F) EACH SIDE	EXTRA @ DISCONT. SUPPORT (E)	EXTRA @ CONT. SUPPORT (D)	NO OF LEGS
1T12	----	----	2
1T12	----	----	2
2T12	2T14	2T14	2

STIRRUPS	
AT SUPPORT	AT MIDSPAN
T8-150	T8-200
T8-150	T8-200
T8-150	T8-200

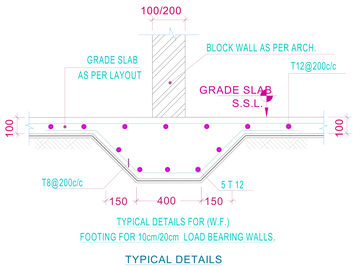
SCHEDULE OF GROUND BEAMS



TYPICAL INTERNAL GROUND BEAM
SC. (1/25)

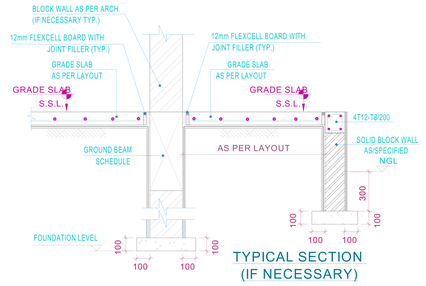


TYPICAL EXTERNAL GROUND BEAM
SC. (1/25)



TYPICAL DETAILS FOR (W.F.)
FOOTING FOR 10cm/20cm LOAD BEARING WALLS.

TYPICAL DETAILS



TYPICAL SECTION
(IF NECESSARY)

TIE BEAM LAYOUT

Govemment Of Ras Al Khaimah
R.A.K. Municipality
Buildings Administration

حكومة رأس الخيمة
دائرة البلدية
إدارة المباني

Approved On — اعتمد في

28 January 2026

Building Permit #: PRMT-26-0000880

Appoved By:

STRUCTURAL NOTES	CONCRETE COVERS :
ACI 318-11	Foundations cover = 75mm
F'C/FCU = 28/35 N/mm2	Columns cover = 40mm
FY = 460N/mm2	Walls cover = 30mm
FYt = 420N/mm2	Beams cover = 40mm
	Slabs & Stairs cover = 25mm

LEGEND:

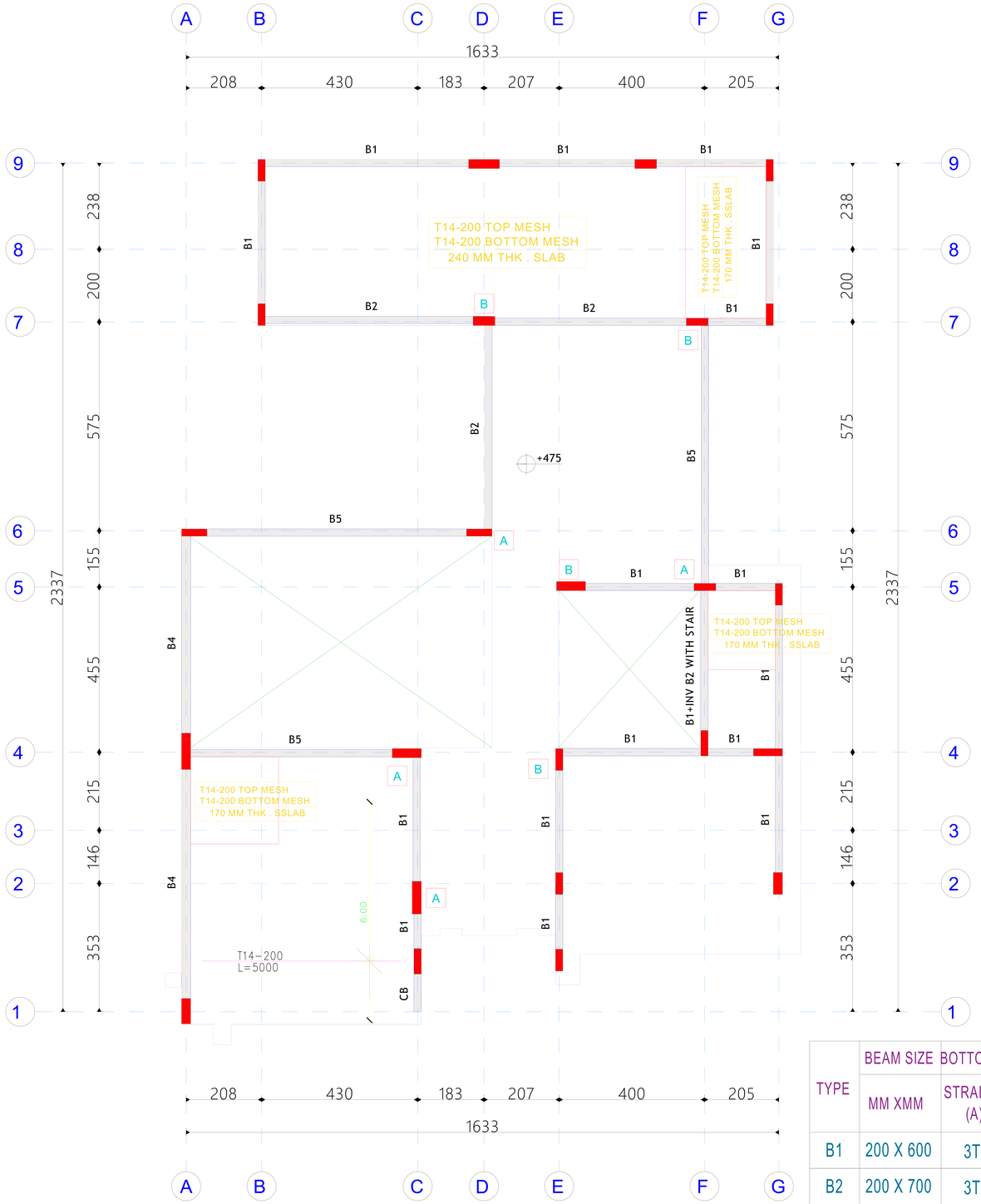
1) COLUMN / WALL TO BE CONTINUED

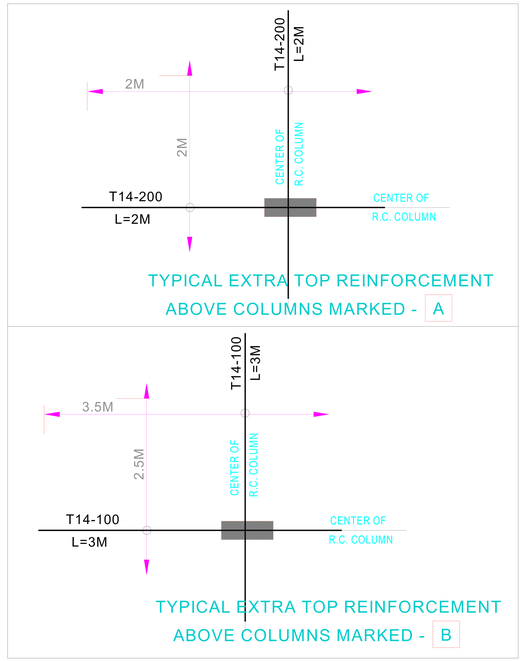
2) COLUMN / WALL TO BE TERMINATED

Savana
Engineering Consultancy & interior design

Client	MAYED MEERZA BASHEER MOHAMMED		
Project	DOUBLE STORY VILLA		
Drg.Title	TIE BEAM LAYOUT		
Area	AWAFI	Floortype	
Case ID.	3136005	Draw. Type.	STRC
Sec No.	1	Com. No.	
Block No.		Plot No.	406131530
Bldg. No.	BLD-26-00570	Floor No.	
Date	1/2026	Sht. No.	S04
Scale	1/100	Job No.	

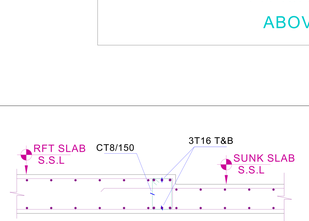
G.Manager	Architect	Designed By
ENG AYMEN	ENG IMEN	ENG IMEN



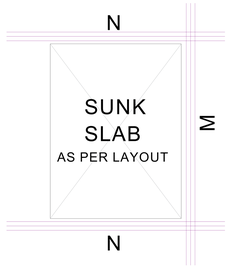


TYPICAL EXTRA TOP REINFORCEMENT
ABOVE COLUMNS MARKED - A

TYPICAL EXTRA TOP REINFORCEMENT
ABOVE COLUMNS MARKED - B



TYPICAL SECTION FOR SUNK SLAB



MARK	RFT	EXTEND
N	3T16 T&B	1000MM
M	3T16 T&B	1000MM

TYPE	BEAM SIZE	BOTTOM REINFORCEMENT		TOP REINFORCEMENT			SIDE BARS O.E.F (F) EACH SIDE	STIRRUPS		NO OF LEGS
	MM XMM	STRAIGHT (A)	EXTRA @ MID SPAN(B)	STRAIGHT (C)	EXTRA @ CONT. SUPPORT (D)	EXTRA @ DISCONT. SUPPORT (E)		AT SUPPORT	AT MIDSPAN	
B1	200 X 600	3T14	1T14	3T14	1T14	1T14	1T12	T8-150	T8-200	2
B2	200 X 700	3T14	2T14	3T14	2T14	2T14	2T12	T8-150	T8-200	2
B3	250 X 700	3T16	2T16	3T16	2T16	2T16	2T12	T8-150	T8-200	2
B4	250 X 800	5T16	----	3T16	2T16	2T16	2T12	T8-150	T8-200	2
B5	250 X 900	6T16	----	3T16	2T16	2T16	2T12	T8-150	T8-200	2
CB	200 X 1000	3T16	----	3T16+2T16			2T12	T8-150	T8-200	2

Govemment Of Ras Al Khaimah
R.A.K. Municipality
Buildings Administration

حكومة رأس الخيمة
دائرة البلدية
إدارة المباني

Approved On — اعتمد في

28 January 2026

Building Permit #: PRMT-26-0000880

Approved By:

LEGEND:

- COLUMN / WALL TO BE CONTINUED
- COLUMN / WALL TO BE TERMINATED

STRUCTURAL NOTES

ACI 318-11
F'C/FCU = 28/35 N/mm2
FY = 460N/mm2
FYt = 420N/mm2

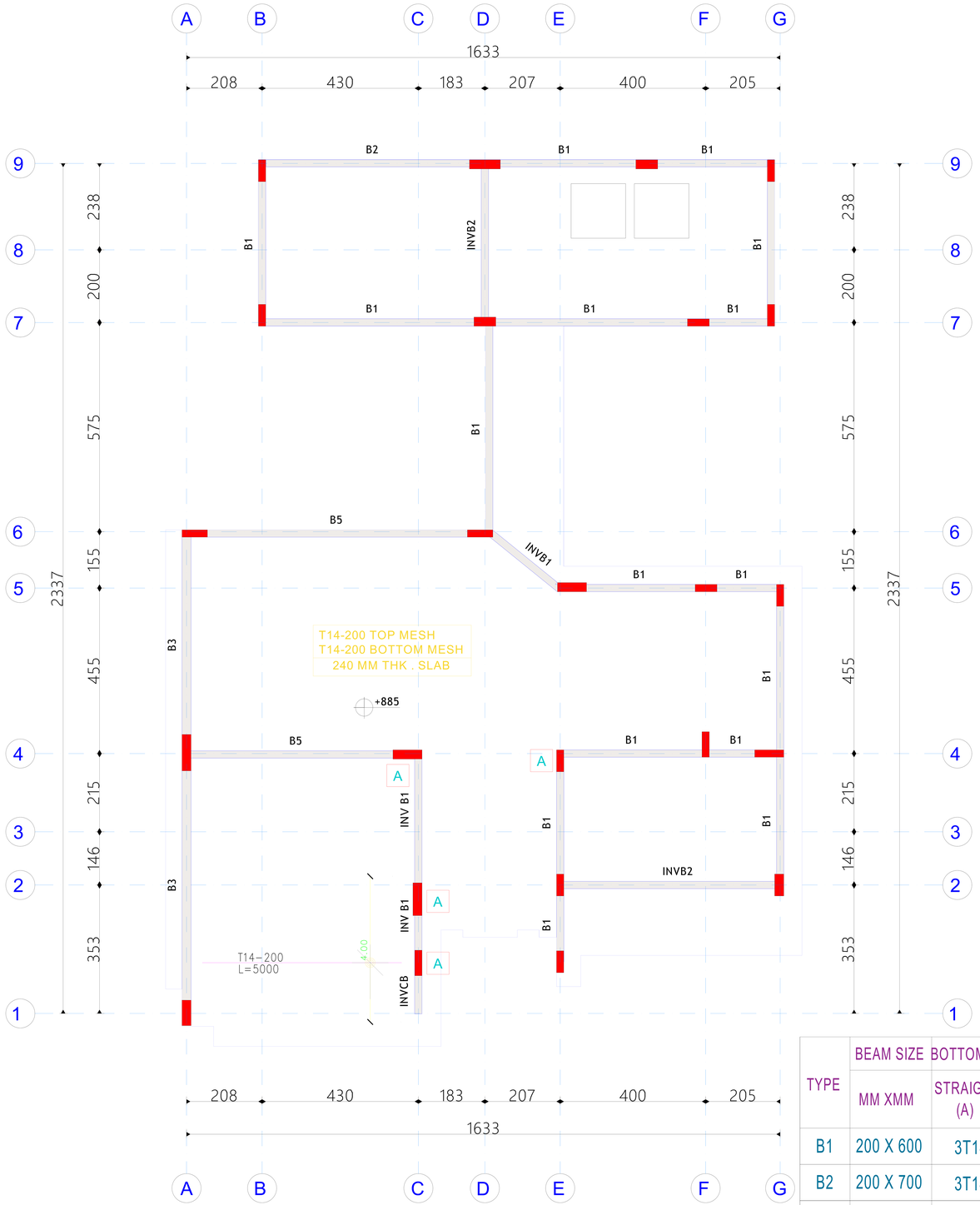
CONCRETE COVERS :

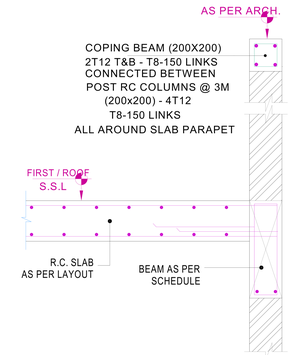
Foundations cover = 75mm
Columns cover = 40mm
Walls cover = 30mm
Beams cover = 40mm
Slabs & Stairs cover = 25mm

Savana
Engineering Consultancy & interior design

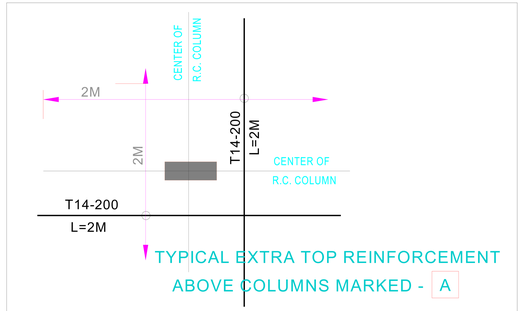
Client	MAYED MEERZA BASHEER MOHAMMED		
Project	DOUBLE STORY VILLA		
Drg.Title	GROUND FLOOR SLAB-BEAMS		
Area	AWAFI	Floortype	
Case ID.	3136005	Draw. Type.	STRC
Sec No.	1	Com. No.	
Block No.		Plot No.	406131530
Bldg. No.	BLD-26-00570	Floor No.	
Date	1/2026	Sht. No.	S05
Scale	1/100	Job No.	

G.Manager	Architect	Designed By
ENG AYMEN	ENG IMEN	ENG IMEN





TYPICAL SECTION FOR PARAPET



TYPICAL EXTRA TOP REINFORCEMENT
ABOVE COLUMNS MARKED - A

TYPE	BEAM SIZE	BOTTOM REINFORCEMENT		TOP REINFORCEMENT			SIDE BARS O.E.F (F) EACH SIDE	STIRRUPS		NO OF LEGS
	MM XMM	STRAIGHT (A)	EXTRA @ MID SPAN(B)	STRAIGHT (C)	EXTRA @ CONT. SUPPORT (D)	EXTRA @ DISCONT. SUPPORT (E)		AT SUPPORT	AT MIDSPAN	
B1	200 X 600	3T14	1T14	3T14	1T14	1T14	1T12	T8-150	T8-200	2
B2	200 X 700	3T14	2T14	3T14	2T14	2T14	2T12	T8-150	T8-200	2
B3	250 X 700	4T16	2T16	3T16	2T16	2T16	2T12	T8-150	T8-200	2
B4	250 X 800	5T16	----	3T16	2T16	2T16	2T12	T8-150	T8-200	2
B5	250 X 900	6T16	----	3T16	2T16	2T16	2T12	T8-150	T8-200	2
CB	200 X 1000	3T16	----	3T16+2T16			2T12	T8-150	T8-200	2

FIRST FLOOR SLAB-BEAMS

Govemment Of Ras Al Khaimah
R.A.K. Municipality
Buildings Administration

حكومة رأس الخيمة
دائـرة البلـدية
إدارة المباني

Approved On — اعتمد في

28 January 2026

Building Permit #: PRMT-26-0000880

Appoved By:

LEGEND:

1) COLUMN / WALL TO BE CONTINUED

2) COLUMN / WALL TO BE TERMINATED

STRUCTURAL NOTES

ACI 318-11
F'C/FCU = 28/35 N/mm2
FY = 460N/mm2
FYt = 420N/mm2

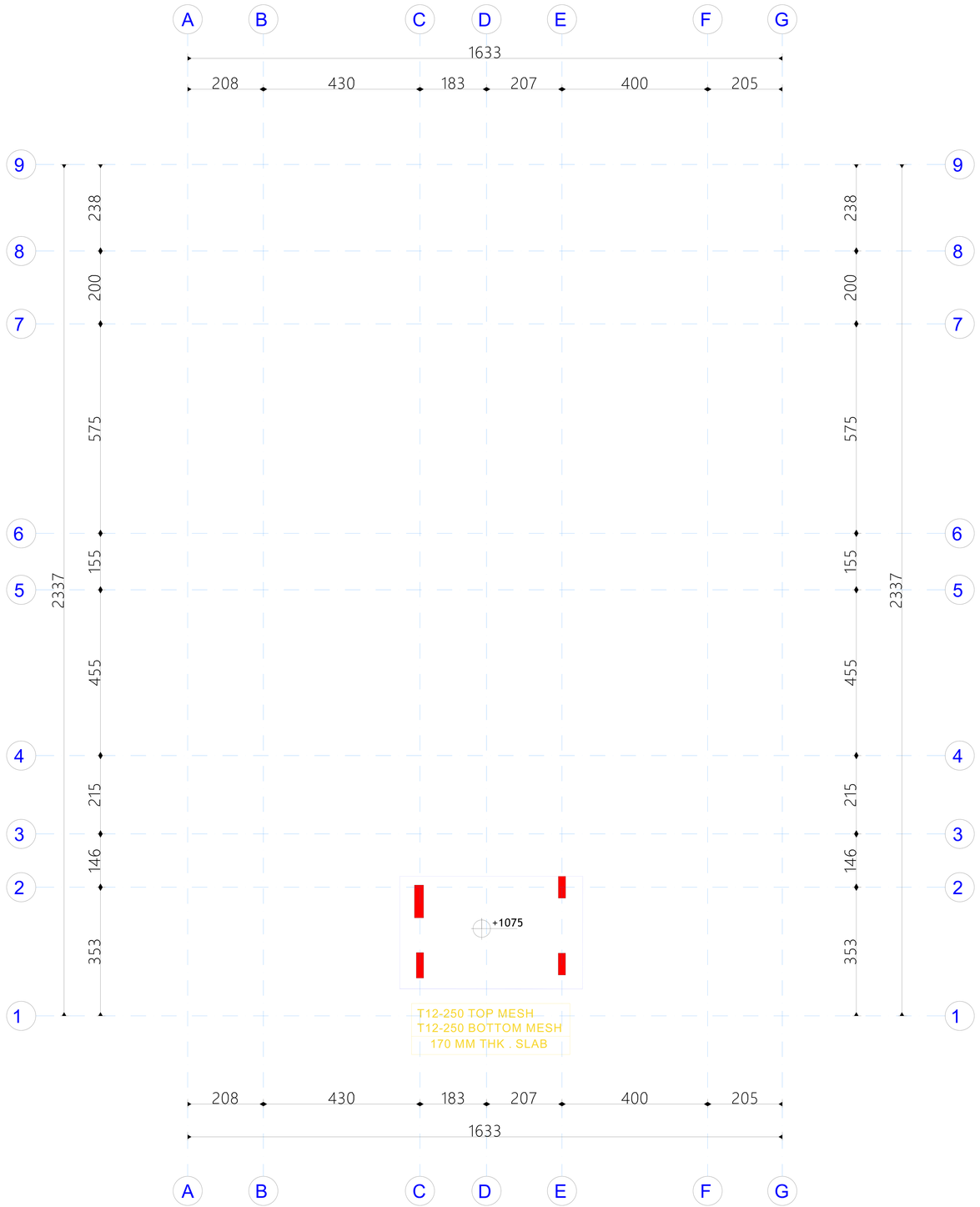
CONCRETE COVERS :

Foundations cover = 75mm
Columns cover = 40mm
Walls cover = 30mm
Beams cover = 40mm
Slabs & Stairs cover = 25mm

Savana
Engineering Consultancy & Interior Design

Client	MAYED MEERZA BASHEER MOHAMMED		
Project	DOUBLE STORY VILLA		
Drg.Title	FIRST FLOOR SLAB-BEAMS		
Area	AWAFI	Floortype	
Case ID.	3136005	Draw. Type.	STRC
Sec No.	1	Com. No.	
Block No.		Plot No.	406131530
Bldg. No.	BLD-26-00570	Floor No.	
Date	1/2026	Sht. No.	S06
Scale	1/100	Job No.	

G.Manager	Architect	Designed By
ENG AYMEN	ENG IMEN	ENG IMEN



Govemment Of Ras Al Khaimah
R.A.K. Municipality
Buildings Administration

حكومة رأس الخيمة
دائرة البلدية
إدارة المباني

Approved On — اعتمد في

28 January 2026

Building Permit #: PRMT-26-0000880

Appoved By:

LEGEND:

1) COLUMN / WALL TO BE CONTINUED

2) COLUMN / WALL TO BE TERMINATED

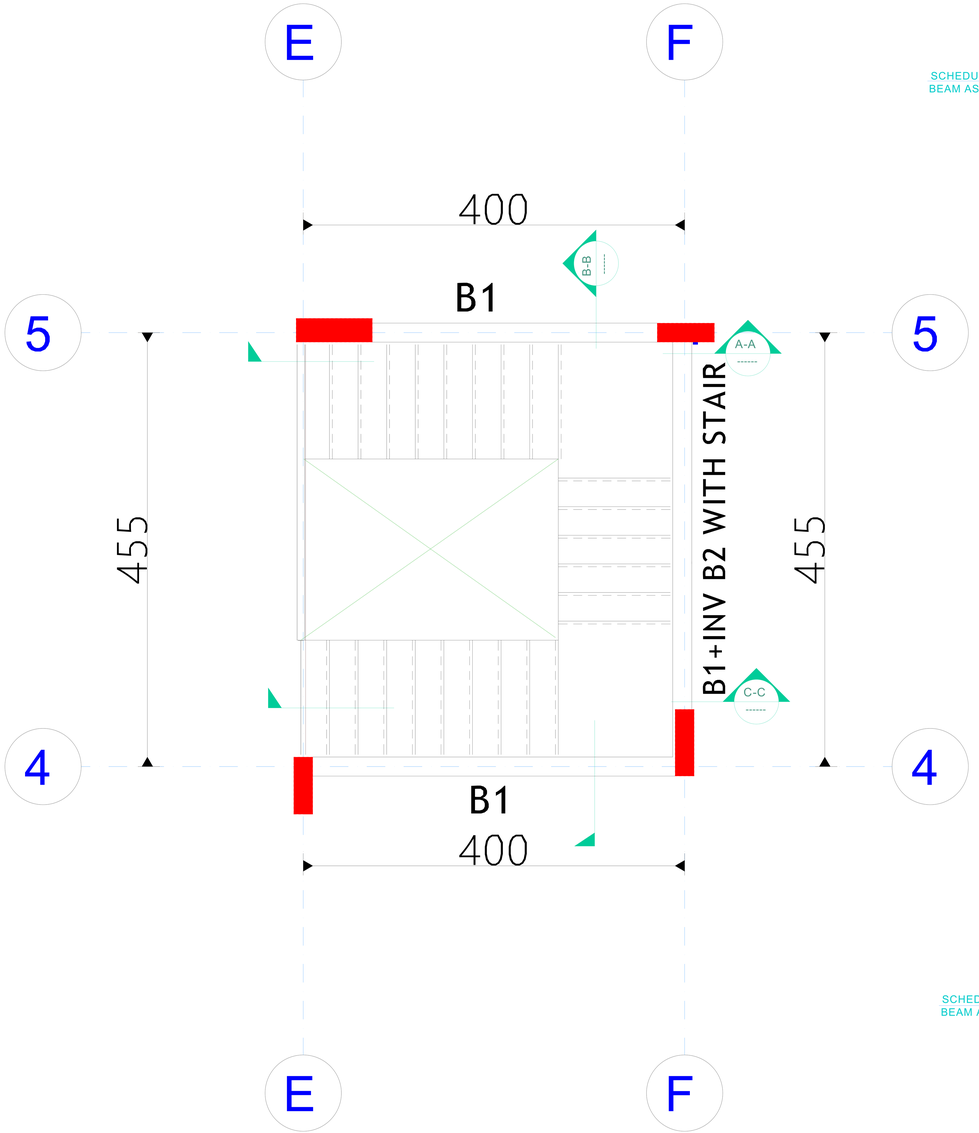
STRUCTURAL NOTES	CONCRETE COVERS :
ACI 318-11	Foundations cover = 75mm
F'C/FCU = 28/35 N/mm2	Columns cover = 40mm
FY = 460N/mm2	Walls cover = 30mm
FYt = 420N/mm2	Beams cover = 40mm
	Slabs & Stairs cover = 25mm

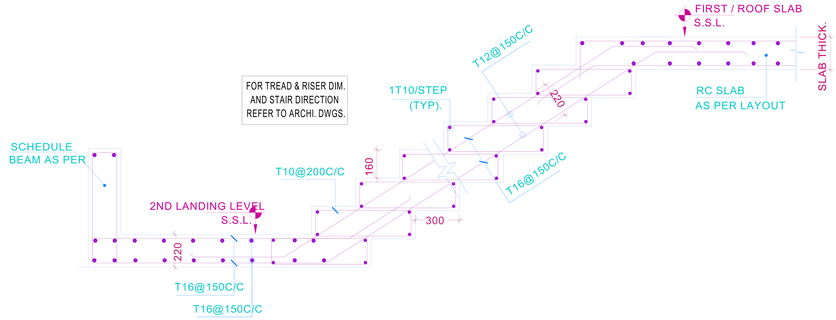
Savana
Engineering Consultancy & interior design

Client	MAYED MEERZA BASHEER MOHAMMED		
Project	DOUBLE STORY VILLA		
Drg.Title	TOP ROOF FLOOR SLAB-BEAMS		
Area	AWAFI	Floortype	
Case ID.	3136005	Draw. Type.	STRC
Sec No.	1	Com. No.	
Block No.		Plot No.	406131530
Bldg. No.	BLD-26-00570	Floor No.	
Date	1/2026	Sht. No.	S06A
Scale	1/100	Job No.	

G.Manager	Architect	Designed By
ENG AYMEN	ENG IMEN	ENG IMEN

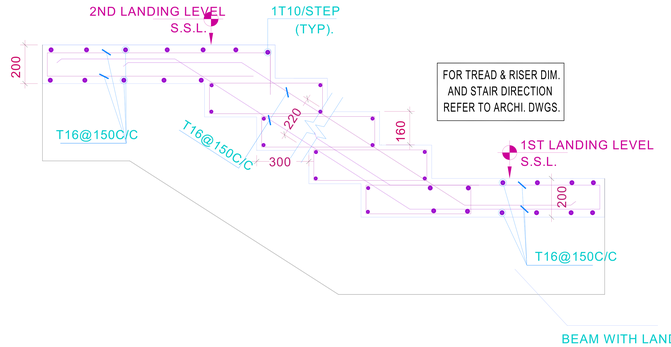
MASTER PLANE





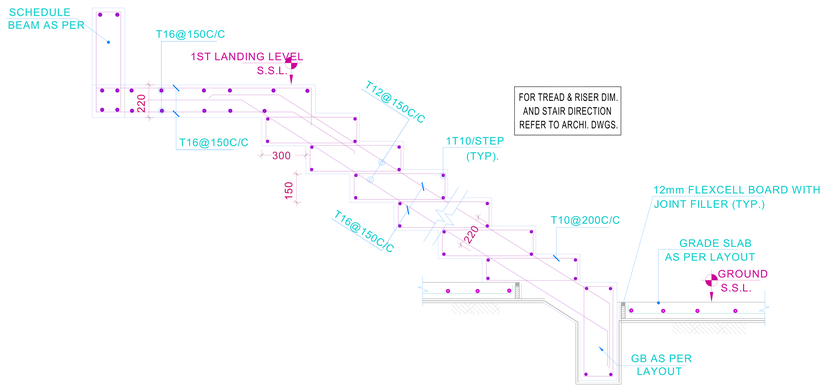
SECTION FROM 2ND LANDING LEVEL TO FIRST / ROOF FLOOR LEVEL

SEC(C-C)



SECTION FROM 1ST LANDING LEVEL TO 2ND LANDING LEVEL

SEC(B-B)



SECTION FROM GROUND FLOOR TO 1ST LANDING LEVEL

SEC(A-A)

STAIR DETAILS

Govemment Of Ras Al Khaimah
R.A.K. Municipality
Buildings Administration

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Savana
Engineering Consultancy & Interior design

Client	MAYED MEERZA BASHEER MOHAMMED		
Project	DOUBLE STORY VILLA		
Drg.Title	STAIR DETAILS		
Area	AWAFI	Floortype	
Case ID.	3136005	Draw. Type.	STRC
Sec No.	1	Com. No.	
Block No.		Plot No.	406131530
Bldg. No.	BLD-26-00570	Floor No.	
Date	1/2026	Sht. No.	S07
Scale	1/100	Job No.	

G.Manager	Architect	Designed By
ENG AYMEN	ENG IMEN	ENG IMEN